Drag reduction from different of shark denticles

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Abstract Each chapter should be preceded by an abstract (no more than 200 words) that summarizes the content.

1 Introduction

Introduce your project! You can perhaps include what led you to choose this topic, or why you chose this certain project idea instead of other similar ideas.

Instead of simply listing headings of different levels we recommend to let every heading be followed by at least a short passage of text. Further on please use the LaTeX automatism for all your cross-references and citations. And please note that the first line of text that follows a heading is not indented, whereas the first lines of all subsequent paragraphs are.

2 Related Work and Background

Use this section to go over any research papers you found useful in your project, or any papers that are relevant to your project topic.

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This is how you [**?**] articles.

3 Data and Methodology

In this study, we investigated the shark denticles by utilizing a computational fluid dynamics simulation in OpenFOAM. We designed the shark denticles inspired by CP scans to use them in the simulation. Firstly, to validate the simulation by replicating case (a) of this paper [reference the paper]. We set up a mesh in OpenFOAM using the snappyHexMesh tool (folder?dictionary?). Because the simulation ran in steady state, we utilized simpleFoam as a solver. One of the main objectives of this study is to create a simulation that is as close to reality as possible. For this reason, we used the Newtonian transport model in transportProperties. Newtonian assumes viscosity stays constant regardless of the shear rate (e.g., water, air) [reference]. –Beğenmedim bunu ya incompressible flow ve fluid density 1 kg/m^{3}*kullandıkdesekdahaiyigibi*.

3.1 Data Collection –Maybe we should change the name???

General description of what is coming- simulation properties and how did we set up paraview. We have data of smooth airfoil (airfoil without denticle), scenario a for [cite paper here] which is a row of denticles located at the L= (id remember the val) on NACA 0012 airfoil with 0 degree angle of attack. We will have more data. Coming...

3.1.1 Simulation Setup

All tecnical drawing of denticle
Measurement of airfoil
Properties of simulation and turbulence, solver etc.

3.1.2 Postprocessing

Talk about how to make calculation or how to use paraview etc.

3.2 Experiment or Methods or idk

Tamam simulasyon falan hazır da nereye konumlandırdın denticleları ya da koyduğun geometriyi nasıl kombinasyonlarda koydun.

3.2.1 Method Subsection 1

sth sth sth –We might not need this one.

4 Results

Provide a description of main findings in chronological order of the data represented in the methods section. Include a summary of research findings. Repeat questions to help readers understand results. Additionally, include a report about information collection, participants, and recruitment.

Show data presented in tables, charts, graphs, and other figures.

Put graphs or picture from the replicated scenario of the reference paper. Put numbers of results Put numbers of comparison Compare and make comments

4.0.1 Validation of simulation

Here will be comparison of the replicated simulation and the reference paper.

4.0.2 Comparison of across denticles

Here will be comparison of different denticle models.

5 Conclusion

Conclusion will be addeed later.

5.1 Applications

Application fields will be added later.

5.2 Concerns and Future Work

Concerns and future work will be added later. s

Acknowledgements We can add an acknowledgement, thank you for Teuscher lab and opportunity that they provided us to have remote research etc. etc.

If you want to include acknowledgments of assistance and the like at the end of an individual chapter please use the `acknowledgement` environment – it will automatically be rendered in line with the preferred layout.

References

1. A. Slavova and V. Ignatov. Edge of chaos in memristor cellular nonlinear networks. *Mathematics*, 10(8):1288, 2022.